Product Specification

AUO Display Plus Corporation

( ) Preliminary Specifications
(■) Final Specifications

Module	**17 inch x 17 inch TFT plate for X-ray sensor**
Model Name	**R1717AS01.3 (92D17170.3R2)**

Customer	**Date**	**Approved by**	**Date**
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Record of Revision

Version	Date (yyyy/m/d)	Page	Old description	New Description
0.1	2022/06/23	All	First draft specification	
0.2	2023/05/08	p.9	• After zero frame, ten dark images (F_1~F_{10}) are acquired continuously with acquisition interval around 1.8sec, which is limited by AUO's testing system.	After zero frame, ten dark images (F_1~F_{10}) are acquired continuously by AUO's testing system.

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1. Operating Precautions

1.1 Electrostatic Discharge Control

1) Remember to avoid ESD (Electro Static Discharge) damage and chemical damage.
2) Static electricity will damage Sensors; please do not touch the Sensor without any grounded device and always keep static electricity under 1KV when handing or assemblage.
3) Management Method of Ionizer: Decay time: < 1KV within 5 sec & balance: +/-100V or less.
4) Make sure of operator wearing Wrist Strap during any Sensor handling process for ESD protection.
5) Less EMI: it will be more safety and less noise.

1.2 Storage

1) All Sensors should be well packed and be stored in a certain moisture and temperature environment.
2) Keep away from water, dust and fire.
3) Keep TFT sensor in dark environment without using or it might have the residual image of light or cause abnormal image when using.

1.3 Usage

1) Please operate Sensor with the correct test voltage/Current/Light source in suitable temperature to avoid any abnormal damage on it. Be sure to turn off the power when connecting or disconnecting the circuit.
2) Do not apply the invalid signal, otherwise, it will cause improper shut down and damage the Sensor.
3) Sudden temperature changes cause condensation. High temperature and humidity may degrade Sensors' performance. Please do not expose the Sensors to the direct powerful light sources, keep away from water/dust/fire, and store them in dark environment.
4) There is no strong vibration or shock or any unsuitable external force in all procedures of buy's sites because it may cause Sensors broken and count to buyer's responsibility.
5) When the panel surface or backside is stained, clean it with air-gun blowing or soft dust-free wiper carefully, please be cautious and do not scratch it.
6) To Take out Sensor from the box should be friendly to avoid damaging the circuit on the sensor. Please avoid the pressure adding to the surface of Sensors
7) Operator is forbidden to use hands to touch the Front side of Sensor. The front Film of Sensor scratches easily, please handle it carefully. Acetic acid or chlorine compounds are not friends with Sensors.
8) In case if a sensor has to be put back into the packing tray after once it was taken out from the container, press at the far ends of the sensor edge softly. Otherwise the TFT sensor may be damaged.
9) There might be a danger to be injured by its fragments when it is broken, please do not touch it by a bare hand.

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2. General Description

This specification applies to the sensor array panel for X-ray sensor R1717AS01.3

2.1 General Specification

ITEMS	Unit	SPECIFICATIONS
Technology		a-Si TFT/PIN diode
Panel size	mm	445.00 (G) x 445.00 (D)
Pixel pitch	um	140
Active area (H x V)	mm	430.08 (G) x 430.08 (D)
Number of pixels		3072 (Gate) x 3072 (Data)

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2.2 Mechanical Specification

Category	ITEMS	Unit	SPECIFICATIONS
Dimension	Panel dimension	mm	445 (G) x 445 (D), tolerance: ±0.2
	Glass substrate	mm	0.7
	Active area dimension	mm	430.08 (G) x 430.08 (D)
	Edge chamfering	mm	0.2±0.1
	Corner cutting	mm	1.5±0.3
	Distance：plate edge – pixel area	mm	Gate pad side: 7.96
		mm	Data pad side: 8.56
		mm	Gate pad opposite side: 6.96
		mm	Data pad opposite side: 6.36
Gate FPC PAD	Pad length	um	1000
	Pad width	um	36
	Pad pitch	um	72
	Number of pad blocks	piece	6
	Number of pads per		562
	Pad block pitch	mm	71.68
	Material		ITO at the top layer
Data FPC PAD	Pad length	um	2200
	Pad width	um	60
	Pad pitch	um	100
	Number of pad blocks	piece	12
	Number of pads per		270
	Pad block pitch	mm	35.84
	Material		ITO at the top layer

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2.3 Electrical Specification

MODEL				STD REVISED	Note
Project Name				R1717AS01.3	
TFT Panel	PIN or not			PIN with	
	Pixel pitch		um	140	
	Panel size		mm	445 (G) x 445 (D)	
	Fill factor (Typ.)		%	65	
	Glass thickness		mm	0.7	
	Number of pixel	gate		3072	
		read out		3072	
	Number of active pixel	gate		3072	
		read out		3072	
TFT	Ron (Typ.)		M ohm	2.2	
			✧ TEG (Avg. of whole testkey) Ron Vds=0.1V, Vg=15V ✧ Sampling item, ratio 1/40		
	TFT Leakage (Max.)		fA/pixel	80	3
PIN	Pixel capacitance (Typ.)	in case of i=1.0um (pF)	pF	1.48	
	Pixel Leakage (Max.)		fA/pixel	3	1.
	Lag: 1st Frame		%	3	2.
Gate line	Resistance (Typ.)		K ohm	6.5	
	Capacitance (Typ.)		pF	149.4	
Data line	Resistance (Typ.)		K ohm	2.4	
	Capacitance (Typ.)		pF	105.9	
Bias line	Resistance (Typ.)		K ohm	3.4	

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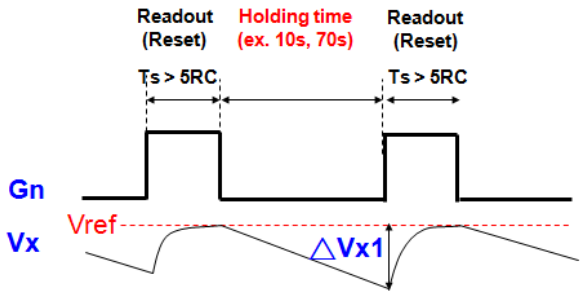
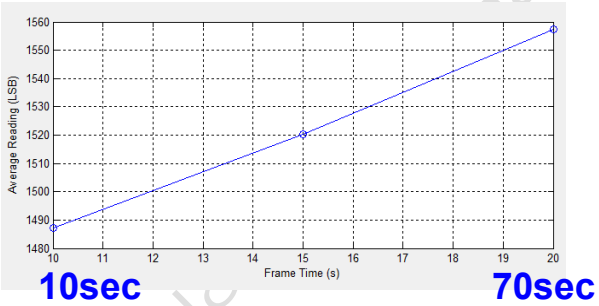
Note1. Definition of Pixel leakage

The value of pixel leakage is to illustrate the pixel dark performance in the sensor array.

- Clean all residual charges in pixels
- Hold in dark environment for 10sec and then capture the image
- Clean all residual charges in pixels, hold in dark environment for 70secs and then capture the image again
- The variation of pixel counts between two holding time is defined as the pixel leakage and can be calculated by the following formula.

$$I = (CNT_{70} - CNT_{10}) \times (e^{-} / LSB) \times 1.6 \times 10^{-19} / \Delta T$$





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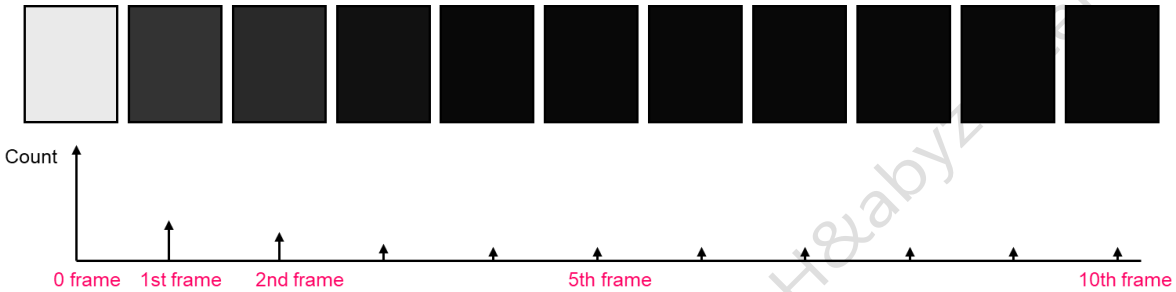
Note2. Definition of Lag

Lag is a residual signal present in image frames subsequent to the frame in which the residual signal was generated.

- Several dark images are acquired until the pixel counts are stable in the dark environment.
- TFT plate is exposed by green light. The dosage of light is tuned to make the pixel value achieving 40000 LSB (60% of 65535 LSB).
- The light image after light exposure is called zero frame (F_0).
- After zero frame, ten dark images (F_1~F_{10}) are acquired continuously by AUO's testing system.
- The lag of each frame is calculated by the formula as below.

$$Lag = \{[Median(F_n) - Offset]/[Median(F_0) - Offset]\} \times 100\%$$



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Note3. Definition of TFT Leakage

The value of TFT leakage is to illustrate the TFT light performance in the sensor array.

- Several dark images are acquired until the pixel counts are stable in the dark environment.
- Set IFS (Integrator fill-scale selection) of AD71124: IDS = 63 (decimal value)
- TFT plate is exposed by strong green light with 10 sec duration and capture light images as Lightimage_10sec.
- Several dark images are acquired until the pixel counts are stable in the dark environment.
- TFT plate is exposed by strong green light with 20 sec duration and capture light image as Lightimage_20sec.
- Subtract two image using (Lightimage_10sec – Lightimage_20sec).
- Calculate Median value of (Lightimage_10sec – Lightimage_20sec) as (CNT10 – CNT20).
- The variation of pixel counts between two holding time is defined as the TFT leakage and can be calculated by the following formula.

$$I = (CNT_{10} - CNT_{20}) \times (e^{-} / LSB) \times 1.6 \times 10^{-19} / \Delta T$$

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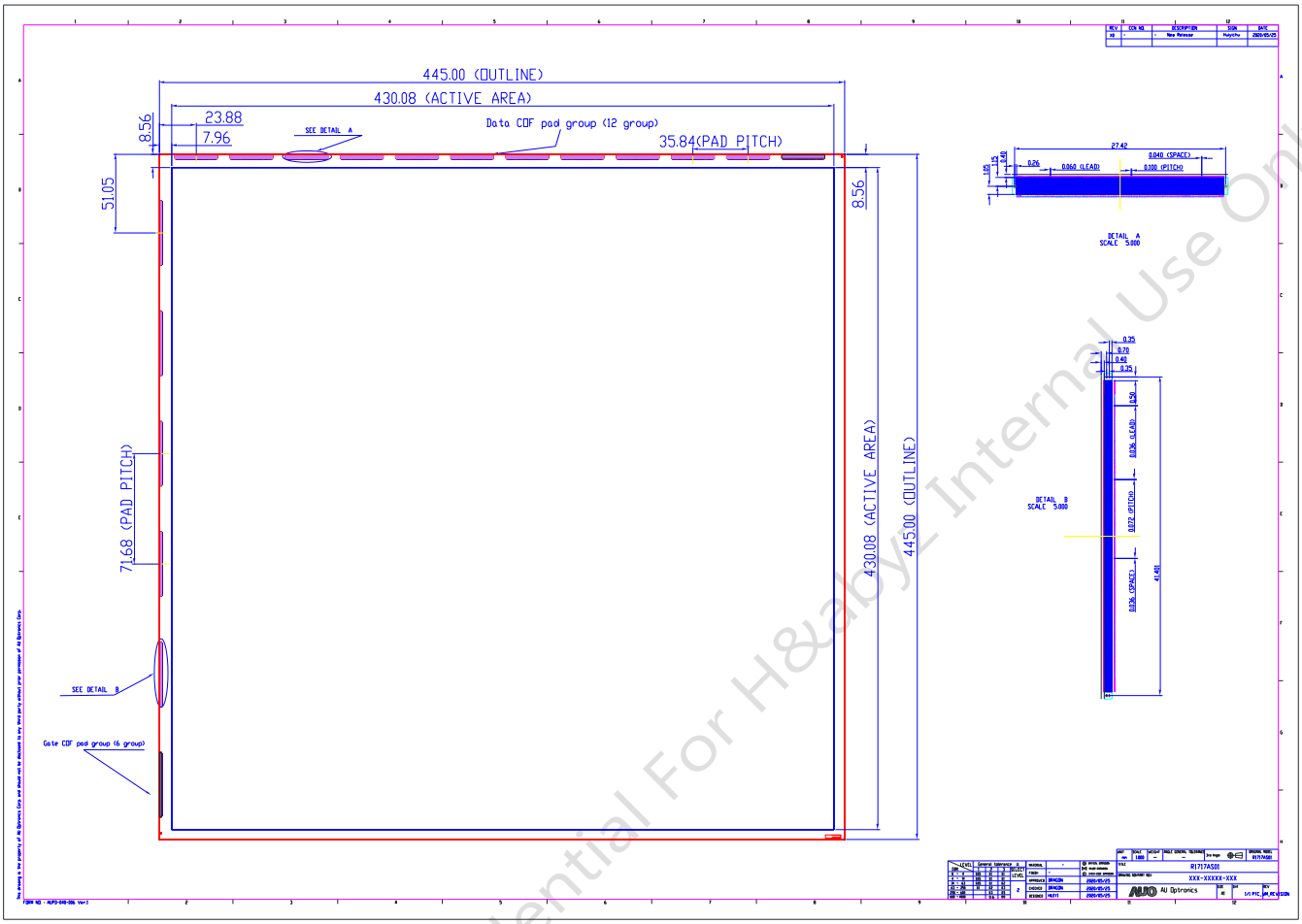
3. Mechanical Characteristics



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4. Pin Out

4.1 Reader FPC

Pin No.	Assignment
1~5	Vcom
6	Dummy
7	Dummy
8~263	Connect to pixel array
264	Dummy
265	Dummy
266~270	Vcom

4.2 Driver

Pin No.	Assignment
1~20	Dummy
21~23	VGL
24~25	Dummy
26~537	Connect to pixel array
538~539	Dummy
540~542	VGL
543~562	Dummy

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5. Reliability Test Criteria

High Temperature High Humidity (HTHH) storage test		Unit	Remark
Temperature	85	degree C	
humidity	85	%RH	
Time	300	h	
Thermal Shock			Remark
-20℃/30 min ,60℃ 30min 100 cycles			

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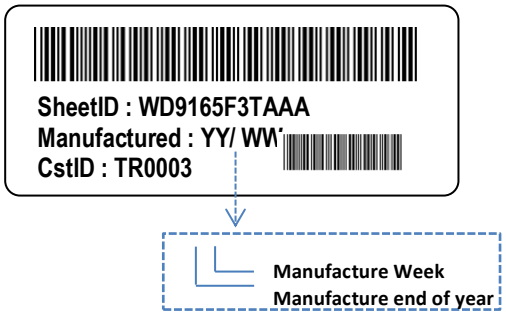
6. Label and Packaging

6.1 Shipping Label

[Label on Black Tray]



[Label on AL bag]

[Label on Carton]

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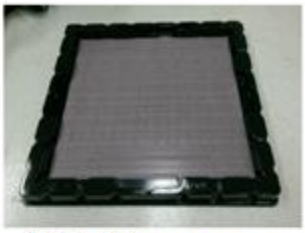
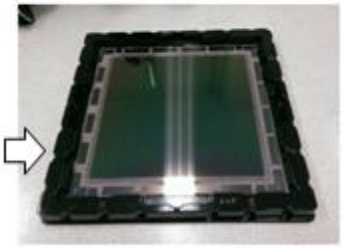
7.2 Carton Package

(1) Pick up PP board

(2) Place bottom tray on PP board



(3) Place EPE spacer on tray



(4) Place X-Ray on EPE spacer, sensor facing up

(5) Place Top tray, and notice location

(6) Stack tray in order, total 5pcs X-ray panel, 5 trays

(7) Fix with creped tape

(8) Place whole stack into ANTI-STATIC bag, plus 2 DESICCANT, then tape the bag

(9) Place ANTI-STATIC bag into AL bag and vacuum it

(10) Put the Al bag into carton

(11) Make sure smooth of Carton

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7.3 Palletizing sequence

- **Pallet size: 1320mm*660mm*140mm**

(1) Fully stacked

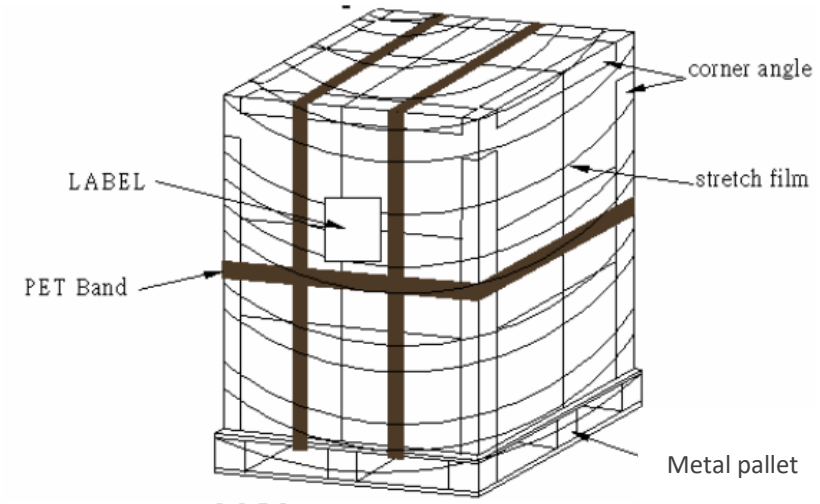
Place cartons between the bottom cardboard on metal pallet. Make sure the cartons don't extend beyond the pallet.

(2) Not fully stacked (One layer total carton numbers>2/3) If not fully stacked, fill with empty cartons until current layer is full then ship as partial pallets.(One layer packing don't use bottom cardboard.)

(3) Not fully stacked (One layer total carton numbers<2/3).

If not fully stacked one layer, must stack to new pallet to place carton one or two rows, use corner pillar and band to fix on pallet, also can refer to method(2).

- **Pallet illustration**



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